UMBANE (UMB) Attested Carbon and Energy Token Summary

by David Robert Lewis

[This is a draft document]

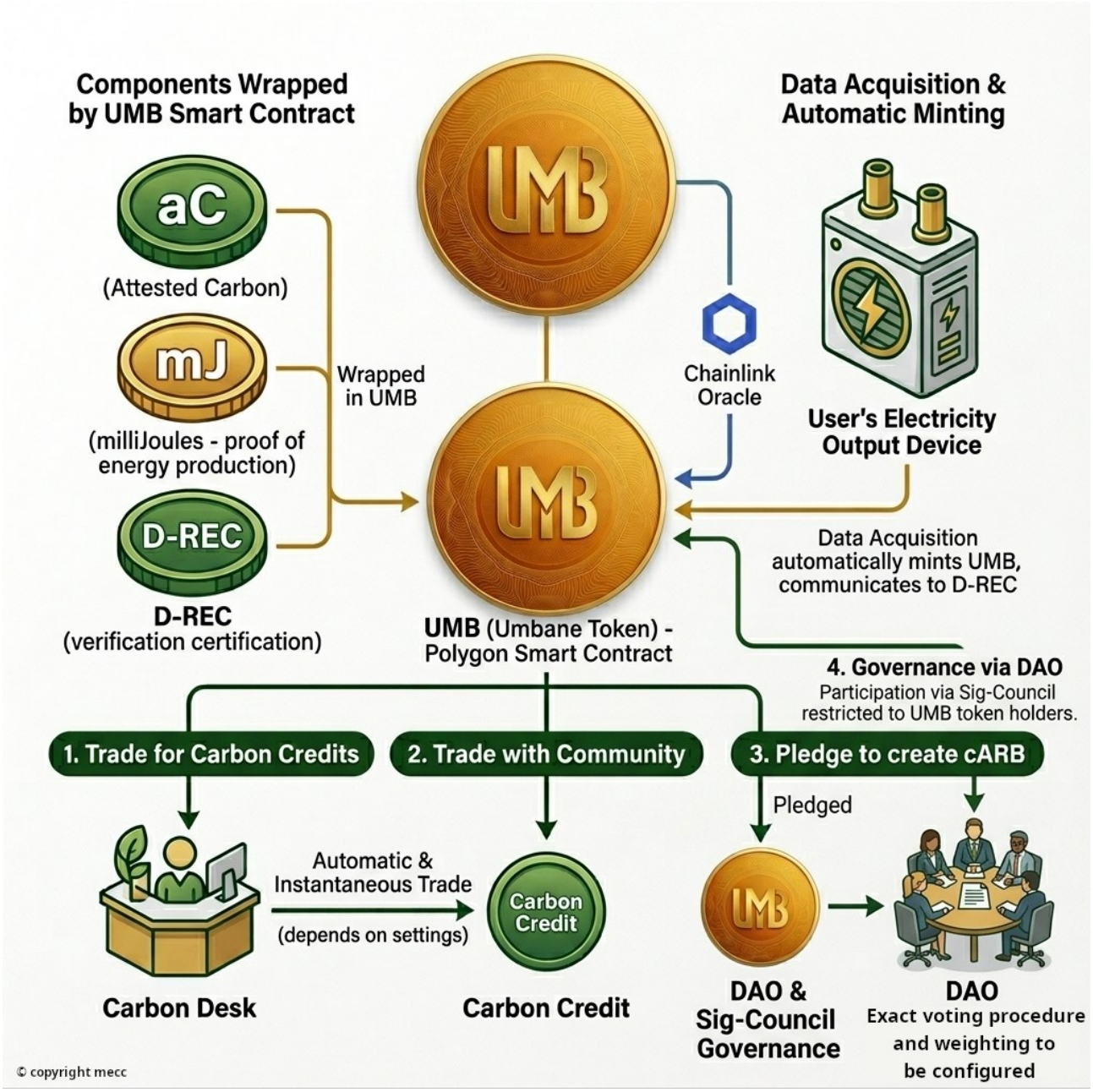
The process begins with the **Generation** phase, where solar panels generate electricity. An IoT device, verified by a Chainlink Oracle, measures this **milliJoule (mJ)** output and transmits the data to a blockchain smart contract.

In the **Verification & Tokenization** phase, the smart contract receives the energy data and creates a digital certificate that includes the **Attested Carbon (aC)** value. It then mints the **UMB Governance Token**, which acts as a standardized EC20 digital asset by wrapping both the raw energy data (mJ) and the certified carbon value (aC) in full compliance with **D-REC** standards.

This leads to the **User Actions & Outcomes** phase, where users can hold UMB to represent their verified renewable energy and participate in various pathways:

- **Voluntary Markets:** They can pledge UMB to receive a **Pledged Carbon Certificate (cARB)**, which is retired for official sustainability claims.
- **Liquidity & Trade:** Users can trade UMB on open markets or within the community.
- **Exchange for Credits:** For automatic pathways, a carbon desk can instantly buy the UMB, facilitating an efficient trade for standard carbon credits.
- **Governance token:** Holders of UMB may participate in governance of the project DAO via the Sig-Council. (We have yet to configure the exact voting procedure and weighting)
- **Fungability:** Holders may trade UMB for stablecoin or other tokens, provide liquidity pools, staking and DeFi via any number of token methodologies.
- **NFT** token can be minted as recognition of a successful carbon retirement.

Image: This visual breakdown captures the entire technical process of the Umbane ecosystem on the Polygon network.



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